TITLE PAGE

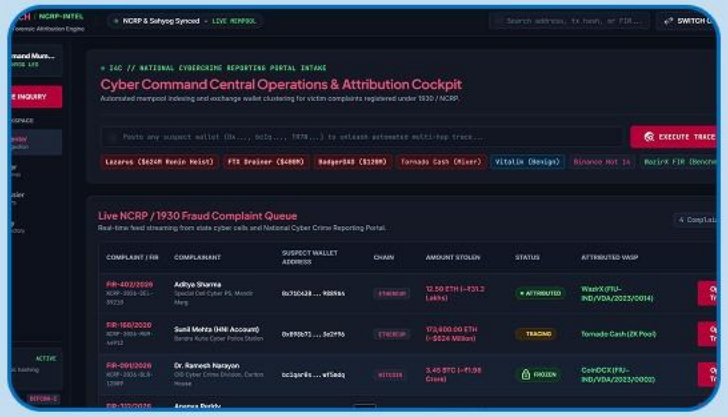
- Problem Statement ID – **SIH26183**
- Problem Statement Title- **Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **184335**
- Team Name - **UnJustice League**

UnJustice League

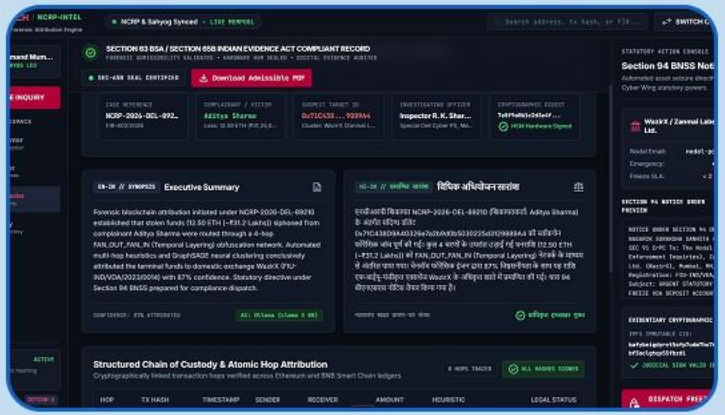
CHAINWATCH

SMART INDIA HACKATHON 2026

CORE SOLUTION: CHAINWATCH, An automated tracking system that ingests victim complaints directly from the national cybercrime helpline (1930) and the official I4C SAHYOG portal, tracking stolen cryptocurrency in under 90 seconds. It automatically generates and sends court-ready "freeze" dossiers to Indian crypto exchanges to lock down funds before criminals can cash out, making these certified files instantly accessible to the investigating officers responsible for the case.







Cyber Command Central Operations:

- Live NCRP queue integration for immediate, one-click trace execution.
- Real-time tracking and continuous cross-referencing against major historical crypto heists.

Interactive Graph Explorer:

- Visually maps complex, multi-hop fund flows across disparate blockchain networks.
- AI-driven node classification achieving a 92.4% attribution accuracy rate.

Automated Evidence Dossier:

- Localized AI compiles legally compliant, sovereign investigative records and chain of custody.
- Instantly generates Section 94 BNSS freeze orders ready for immediate exchange dispatch.

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TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

TECH STACK

CHAINWATCH sovereign forensics architecture

01 / FRONTEND

Next.js · React · Tailwind CSS · D3.js · Deck.gl · TypeScript

02 / BACKEND & STREAMING

FastAPI · Apache Kafka · Apache Flink · Redis · Meilisearch

03 / ML & GRAPH INTELLIGENCE

PyTorch Geometric · Apache AGE PostgreSQL · Ollama / Llama 3 · NetworkX · NumPy

04 / BLOCKCHAIN & RPC

Subsquid · Bitcoin Core · Web3.py / Python · Multi-Chain: Ethereum, BSC, Tron

05 / INFRA & EVIDENCE

Typst / court PDF reports · OpenSSL / SHA-256 · Docker & K3s · Grafana & Prometheus

Continuous Case Monitoring Loop

01 / INGESTION

NCRP / 1930 Portal
Complaint webhook intake

Blockchain RPCs
ETH · BSC · TRON nodes

Queue & Cache
Redis + Celery workers

API Gateway
FastAPI ingestion layer

02 / GRAPH ENGINE — 47.2s avg

BFS Traversal
Hop-by-hop graph walk

GraphSAGE GNN
PyTorch Geometric · node embeddings

Graph Database
Neo4j · relationship store

Heuristic Clustering
Peel chains · deposit reuse

Re-evaluation Loop

03 / EVIDENCE

VASP Attribution
FIU-IND registered exchanges

SHA-256 Sealing
Tamper-proof evidence hash

LLM Narrative
Llama-3 local · case summary

Dossier Compiler
ReportLab · legal PDF output

04 / DISPATCH

IO Dashboard
Next.js 14 · D3.js graph view

Sec 94 BNSS Freeze
Statutory freeze notice

SAHYOG Dispatch
MHA inter-agency API

Case Status Update

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FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

TECHNICAL
Infrastructure & Sovereignty

NIC MeghRaj → ~10GB pruned nodes → Local LLM inference → Zero proprietary licenses

✓ NIC MeghRaj ✓ ~10GB nodes ✓ Local LLM ✓ No licenses

OPERATIONAL
IO-Centric Workflow

NCRP webhook → Graph triage → Auto Section 94 freeze notice

✓ NCRP webhook ✓ Graph triage ✓ Auto notice

REGULATORY
Evidentiary Admissibility

Sec 63 BSA → Sec 94/107 BNSS → FIU-IND → Compliant evidence / freeze workflow

✓ Sec 63 BSA ✓ Sec 94/107 BNSS ✓ FIU-IND

ECONOMIC
Procurement Feasibility

No foreign subscriptions → Open-source maintenance → Lower recurring costs

Cost to trace: ₹0.50 per trace

✓ No foreign currency ✓ Open source ✓ Lower costs

RISKS

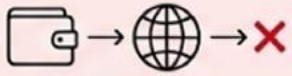
Mixer Obfuscation — HIGH

Trail breaks at tumbler. BFS blind beyond mixer.



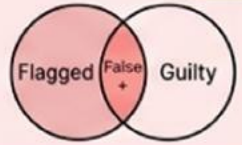
Offshore DEX Blind Spots — HIGH

No KYC/VASP data. Attribution stops at foreign exchange.



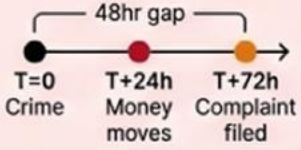
GNN Misclassification — MED

False positives may flag innocent wallets.



Delayed Complaint Reporting — LOW

Funds laundered before NCRP complaint is filed.



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IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

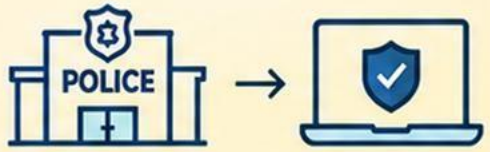
IMPACT

CRIPPLING THE LAUNDERING PIPELINE



Chokes the primary crypto exit-ramp used by international cyber syndicates.

DECENTRALIZING CYBER DEFENSE



Brings enterprise-grade forensic capabilities to local police stations, reducing dependence on elite cyber units.

DETERRING FUTURE SYNDICATES



90-second automated tracing increases the risk of detection and can deter criminal networks targeting Indian citizens.

RESTORING DIGITAL TRUST



Creates a sovereign safety net so complax digital financial crimes end up tracked with transparent evidence.

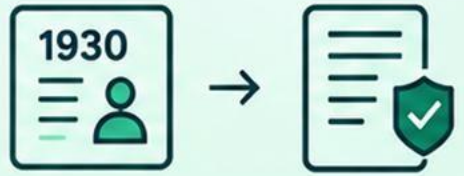
BENEFITS

GOLDEN HOUR RECOVERIES



Cuts investigation time from 72 hours to 47 seconds, helping intercept stolen funds before permanent laundering.

INSTANT LEGAL ACTION



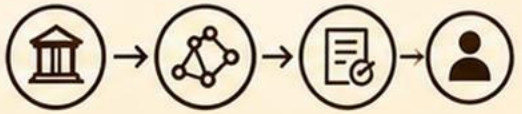
Converts a citizen's 1930 portal complaint into an auto-drafted statutory freeze notice sent to the exchange.

SWIFT VICTIM RESTITUTION



Plain-language legal reports accelerate court proceedings and help victims access recovered funds sooner.

TRANSPARENT CASE PROGRESS



Explainable, court-admissible evidence chains show victims where their money moved instead of relying on black-box guesses.

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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

MAIN REPOSITORY

https://github.com/p4rthh/sih-blockchain-prototype

Prototype Video Link

https://www.youtube.com/watch?v=puK3wHBqn8s

ACADEMIC & LITERATURE REFERENCES

1. Hamilton, W., Ying, Z., & Leskovec, J. (2017). **Inductive representation learning on large graphs.** Advances in Neural Information Processing Systems (NeurIPS), 30, 1024–1034.

2. Weber, M., Chen, J., Xu, B., & Yilmaz, A. (2019). **Anti-money laundering in Bitcoin: Experimenting with Graph Neural Networks for financial forensics.** IEEE International Conference on Data Mining (ICDM), 1222–1227.

3. Victor, F., & Lüders, F. (2021). **Detecting money laundering patterns on EVM-based blockchains using graph-based temporal heuristics.** IEEE Transactions on Network and Service Management, 18(3), 3102–3115.

DATASETS & DATA SOURCES

SAHYOG & NCRP Portals: Ministry of Home Affairs / Indian Cyber Crime Coordination Centre (I4C), "National Cyber Crime Reporting Portal (NCRP) & SAHYOG Complaint Feeds," [Online]. Available: https://cybercrime.gov.in/

Subsquid EVM Data Lake: Subsquid Network, "Decentralized On-Chain Indexing Data Lake," [Online]. Available: https://docs.subsquid.io/

Etherscan & Wallet Directories: Etherscan, "Etherscan Label Cloud & Known Exchange Wallet Registry," [Online]. Available: https://etherscan.io/labelcloud

Arkham Intelligence: Arkham, "On-Chain Entity Intelligence & VASP Attribution Database," [Online]. Available: https://www.arkhamintelligence.com/

OpenRPC Providers: dRPC, "Decentralized OpenRPC Multi-Chain Interface & Endpoint Aggregator," [Online]. Available: https://drpc.org/

CORE FRAMEWORKS & TOOLS

ML/AI

PyTorch Geometric (PyG), GraphSAGE, Llama (via Ollama), Apache AGE (PostgreSQL Graph Extension), Flink CEP

Blockchain Indexing & Tracing

Subsquid Indexer (EVM Data Lake), dRPC / OpenRPC Protocol, Bitcoin Electrum Protocol, Apache AGE Cypher Queries

Backend & Streaming

Python, FastAPI, Apache Kafka, Apache Flink, Async Redis

Frontend & Reporting

Next.js, D3.js (Force Graph), Deck.gl, Typst (PDF Engine)

DOCUMENTATION

Kafka
https://kafka.apache.org/43/getting-started/introduction/

Ollama
https://docs.ollama.com/api/introduction

Redis
https://redis.io/docs/latest/

Next.js
http://nextjs.org/docs